🔷 High-level overview (start here)

This project is an Annotation Transfer Tool that significantly reduces manual labeling effort by automatically transferring annotations from previously human-labeled images to new images using deep image similarity.

In short:

👉 **Find similar images → reuse their annotations → verify visually → export back to CVAT**

🔷 What problem does it solve?

Manual annotation is slow and expensive. Even auto-annotation models struggle when data is limited or domain-specific. This tool leverages already annotated images and transfers their labels to new images by identifying visually similar samples using deep learning.

🔷 Core idea (very important)

Instead of predicting bounding boxes from scratch, the tool searches for visually similar images using deep feature embeddings and transfers annotations from the closest matches.

This makes it:

- Faster than manual labeling
- More reliable than blind auto-annotation
- Especially effective in **low-variation industrial datasets**

🔷 End-to-end pipeline explanation

1️⃣ Configuration-driven setup

The entire pipeline is controlled through a single config.yml, which defines input paths, S3 locations, and processing parameters.

This makes the pipeline:

- Reproducible
- Easy to reconfigure across datasets

2️⃣ Image & annotation ingestion

Images are downloaded from S3 based on a Datumaro JSON file exported from CVAT. Only relevant images are pulled, avoiding unnecessary data transfer.

Scripts involved:

- s3_download.py
- datuamaru_crop_sort.py

3️⃣ Label normalization

Annotations are normalized into a consistent format so that downstream processes like cropping and similarity matching work reliably.

Script:

- normalize_labels.py

4️⃣ Crop, mask, and label generation

For each annotated object, the tool generates image crops, corresponding label files, and binary masks.
This creates a **reference library of labeled object appearances**.
Script:

- crops_generation.py

5️⃣ Deep image similarity search (core intelligence)

A deep learning–based feature extractor generates embeddings for image crops. Using DeepImageSearch, the tool identifies visually similar images across the dataset.
Script:

- CheckSimilarityScore_gpu.py

This is where:

- Feature-based search replaces brute-force labeling
- Previously labeled examples guide new annotations

6️⃣ Annotation transfer & correction

Annotations from the most similar matches are transferred to the target images. Any mismatches or incorrect labels are corrected and mapped back to the original annotation files.
Script:

- Correcting_matching.py

This step ensures:

- Label consistency
- Reduced noise from similarity errors

7️⃣ Visualization for validation

Both normalized and denormalized annotations are visualized to verify correctness before final export.
Scripts:

- visualization_normalized.py
- visualization_denormalized.py

This acts as a **human-in-the-loop sanity check**.

8️⃣ Export back to CVAT

Finally, the updated annotations are converted back into Datumaro JSON format, making them directly uploadable to CVAT.
Script:

- fetch_json_data.py

🔷 Two ways to run the system

✔ Full automation

The entire pipeline can be run with a single command (run.py) for batch processing.

✔ Step-by-step execution

Each stage can also be executed independently for debugging, validation, or partial runs.

🔷 Interactive demo

A Jupyter notebook (DEMO.ipynb) demonstrates the entire workflow interactively, making it easy to understand intermediate outputs and debug results.

🔷 One-liner summary (gold for interviews)

I built a deep similarity–based annotation transfer system that automatically reuses existing human annotations to label new images, reducing manual labeling effort while maintaining annotation quality.